V.L Memorial Public School

475 Vijay Nagar

CHAPTER-4-QUESTIONS-CLASS7TH-SCIENCE

Class 07 - Science (Curiosity)

Section A

1. Which of the following metals is the least reactive? **[1]**

   a) Gold
   b) Magnesium
   c) Iron
   d) Potassium

2. Which metal is used to make the filaments of electric bulbs? **[1]**

   a) Iron
   b) Copper
   c) Aluminium
   d) Tungsten

3. Which of the following non-metals is essential for life? **[1]**

   a) Nitrogen
   b) Carbon
   c) Oxygen
   d) Chlorine

4. Why are metals commonly used for cooking vessels? **[1]**

   a) They conduct heat well
   b) They block heat flow
   c) They prevent heat transfer
   d) They break into pieces

5. Which pair contains materials that can be beaten into thin sheets? **[1]**

   a) Copper and aluminium
   b) Coal and wood
   c) Wood and sulfur
   d) Coal and sulfur

6. Which material is a poor conductor of heat? **[1]**

   a) Wood
   b) Iron
   c) Aluminium
   d) Copper

7. Which of the following is not a property of metals? **[1]**

   a) Brittleness
   b) Sonority
   c) Ductility
   d) Malleability

8. Which combination describes common properties of metals? **[1]**

   a) Brittle, dull, non-conducting
   b) Soft, brittle, non-lustrous
   c) Hard, lustrous, ductile
   d) Non-lustrous, brittle, insulating

9. Which of the following non-metals is used for making matchsticks? **[1]**

   a) Sulphur
   b) Nitrogen
   c) Phosphorus
   d) Carbon

10. Which of the following is an example of a compound formed between a metal and a **[1]**

non-metal?

a) Water
b) Sodium chloride
c) Oxygen
d) Hydrogen

11. What is gradual deterioration of metal surfaces called? **[1]**

a) Polishing of metals
b) Galvanisation of metals
c) Rusting of metals
d) Corrosion of metals

12. Which non-metal is used in manufacturing fertilisers? **[1]**

a) Chlorine
b) Nitrogen
c) Carbon
d) Oxygen

13. Which of the following is a property of non-metals? **[1]**

a) Sonorous
b) Malleable
c) Good conductor of electricity
d) Dull and brittle

14. Which of the following is a metal? **[1]**

a) Oxygen
b) Carbon
c) Iron
d) Sulphur

15. Which of the following is a property of both metals and non-metals? **[1]**

a) They are both shiny
b) They form alloys
c) They both form ionic compounds
d) They are both malleable

16. The nature of metal oxides is **[1]**

a) neutral
b) basic
c) all of these
d) acidic

17. Which property is characteristic of metals? **[1]**

a) High melting and boiling points
b) Dull in appearance
c) Poor conductors of heat
d) Poor conductors of electricity

18. Which property allows metals to be drawn into wires? **[1]**

a) Malleability
b) Ductility
c) Sonority
d) Brittleness

19. Which property explains the use of metal wires in stringed musical instruments? **[1]**

a) Ductility
b) Brittleness
c) Sonority
d) Malleability

20. Which ancient iron structure in India is famous for its resistance to rust? **[1]**

a) Pashupatinath Temple
b) Iron Pillar
c) Kaiyuan Temple
d) Ananda Temple

21. **Assertion (A):** Copper is used in electrical wiring. **[1]**
Reason (R): It is ductile and a good conductor.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Iron nails get coated with a reddish-brown layer when exposed to moisture. **[1]**
Reason (R): Iron undergoes rusting due to oxidation.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Phosphorus is stored underwater. **[1]**
Reason (R): It catches fire when exposed to air.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Non-metals are non-sonorous. **[1]**
Reason (R): Non-metals produce a ringing sound when struck.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** Sulphur is a non-metal used in matchsticks. **[1]**
Reason (R): It is highly malleable.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. **Assertion (A):** Non-metals are poor conductors of heat. **[1]**
Reason (R): Non-metals lack free electrons for heat transfer.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

27. **Assertion (A):** Sodium is stored in kerosene oil. **[1]**
Reason (R): Sodium reacts very slowly with oxygen and water.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

28. Which non-metal is a key component of proteins and carbohydrates? **[1]**
29. What is the physical state of most metals at room temperature? **[1]**
30. Solution of which non-metal is applied on wounds as an antiseptic? **[1]**
31. How many elements are currently known? **[1]**

32. Describe the importance of recycling iron and aluminium in India. **[1]**

33. Which non-metal is essential for breathing? **[1]**

34. What are rust and rusting in relation to iron objects? **[1]**

Section B

35. Match the following: **[2]**

Column A	Column B
(a) Copper	(i) Non-metal
(b) Sulphur	(ii) Metal
(c) Iron	(iii) Dull and brittle
(d) Coal	(iv) Lustrous and metallic

36. Match the following: **[2]**

Column A	Column B
(a) Sulphur	(i) Used in fertilisers
(b) Carbon	(ii) Essential for respiration
(c) Nitrogen	(iii) Used in making antiseptic creams
(d) Oxygen	(iv) Found in fuels and living things

37. What happens when oxygen reacts with magnesium and sulphur. What are the main differences in the nature of products formed? **[2]**

38. Why do iron objects rust faster in coastal areas? **[2]**

39. Explain why gold is used in making jewelry. **[2]**

40. Why is phosphorus stored underwater? **[2]**

41. Explain the meaning of corrosion with suitable examples. **[2]**

42. Explain why aluminium is used for making cooking utensils. **[2]**

43. Explain how non-metals react with water and oxygen. **[2]**

Section C

44. Why do non-metals not conduct electricity, while metals do? **[3]**

45. Why aluminium is used for wrapping of food items? **[3]**

46. Why does a metal spoon become hotter than a wooden spoon in the same hot water? **[3]**

47. List any four physical properties of metals. **[3]**

48. A metal 'X' is used in making foils for food packaging. Identify 'X' and give a reason for its use. **[3]**

49. Suggest a method to prove that a given material is a metal or non-metal. **[3]**

50. Why do some metals lose their shine over time? **[3]**

Section D

51. **Read the following text carefully and answer the questions that follow:** **[4]**

A piece of coal or a lump of sulfur does not show this behaviour. They break into pieces and are said to be brittle. On the other hand, wood neither gets flattened into a sheet nor breaks into pieces. Therefore, wood is neither malleable nor brittle. Some ornaments, like bangles, necklaces, earrings, etc., are also made from metal wires.

Metal wires play important functions in a variety of stringed musical instruments, such as veena, sitar, violin, and guitar. This property of materials by which they can be drawn into wires is called ductility. This property of ductility is mainly possessed by metals. The ductile nature of metals enables for the creation of the tea strainer with metal wire. The metal spoon and the metal coin produce a ringing sound. Coal and wood, on the other hand, produce dull sounds. This property of metals that enables them to produce a ringing sound is called sonority, and metals are said to be sonorous in nature. The transfer of heat from one point to another of a material is called conduction, and materials that transfer heat are called conductors. So we can say that metals are good conductors of heat, whereas wood is a poor conductor of heat. Due to this mostly metal vessels are used for cooking, and their handles are made with wood or other materials that do not conduct heat.

Question:

1. What is the property of metals that allows them to be drawn into wires? (1)
2. Why are the handles of cooking vessels usually made of wood? (1)
3. Explain the difference between malleability and brittleness using coal, sulfur, and wood as examples. (2)

   OR

   What is sonority? Name two objects that show this property. (2)

52. **Read the following text carefully and answer the questions that follow:** **[4]**

Substances like sulfur and phosphorus behave differently with air and water than metals. Phosphorus is stored in water as it catches fire when exposed to atmospheric air. These substances are usually soft and dull in appearance. They are neither malleable nor ductile, and they are not sonorous. They are also poor conductors of heat and electricity. These are called non-metals. Their oxides are acidic in nature. Some other non-metals are oxygen, hydrogen, nitrogen, carbon, etc. These must not be confused with materials such as plastic, glass, wood, rubber, and paper. These materials are not classified as metals or non-metals because they are not elements.

Questions

a. Why is phosphorus stored in water? (1)
b. Are non-metals good conductors of heat and electricity? (1)
c. Write any two physical properties of non-metals. (2)

   OR

   Why are plastic, wood, and rubber not considered non-metals? (2)

53. **Read the following text carefully and answer the questions that follow:** **[4]**

Moist air is responsible for the development of brown deposits on objects made of iron. This brown deposit is called rust. The process of formation of rust on objects made of iron is called rusting. Many other metals also exhibit discolouration when kept open in the atmosphere. Gradual deterioration of metal surfaces caused by air, water, or other substances is known as corrosion. Every year, an enormous amount of money is spent to replace or repair iron structures damaged due to rusting. The rusting of iron can be prevented by several methods such as painting, oiling, greasing,

and applying a protective layer of zinc metal on iron (galvanisation). The Iron Pillar of Delhi was made in the time of Chandragupta II more than 1600 years ago. It is about 8 metres high and weighs more than 6000 kilograms. What is interesting about this iron pillar is that despite all these years of facing winds, rainfall, and intense weather, it has barely any rust.

Questions:

a. What is the brown deposit formed on iron objects called? (1)

b. Name any one method used to prevent rusting of iron. (1)

c. What is corrosion? Give one example. (2)

OR

What makes the Iron Pillar of Delhi so remarkable? (2)

Section E

54. Explain how the reactivity series helps in understanding metal reactions. **[5]**

55. Analyze the contribution of iron to the progress of civilisation in India. **[5]**

56. Why is gold preferred for making jewelry? Explain with properties. **[5]**

57. Evaluate how the different properties of metals contribute to their varied practical uses. **[5]**

58. Explain the differences between metals and non-metals based on their physical properties. **[5]**

Section F

59. **Fill in the blanks:** **[7]**

(a) Non-metals are _____ conductors of heat and electricity. **[1]**

(b) Non-metals are generally _____ in appearance. **[1]**

(c) Non-metals like _____ do not conduct electricity. **[1]**

(d) The shine on the surface of metals is known as _____. **[1]**

(e) Charcoal and sulphur are examples of _____ because they break when beaten. **[1]**

(f) _____ is the only metal that is liquid at room temperature. **[1]**

(g) The only metal that is liquid at room temperature is _____. **[1]**

Section G

60. **State whether the given statement is True or False:** **[9]**

(a) Scientists can design new materials by changing the proportion of metals and non-metals. **[1]**

(b) Non-metals are good conductors of electricity like metals. **[1]**

(c) Mercury is the only metal that is liquid at room temperature. **[1]**

(d) Metals are generally good conductors of heat and electricity. **[1]**

(e) A combination of metals and non-metals is useful in making alloys with improved properties. **[1]**

(f) We can use wood as an alternative to copper wires in circuits. **[1]**

(g) Rusting only occurs in the presence of water. **[1]**

(h) Aluminium foil is used in packaging food because it is a good conductor of heat. **[1]**

(i) Non-metallic oxides turn red litmus paper blue. **[1]**